

ELE228 - Programming for Circuits and Systems Preliminary Work 3

1 Questions

1. Consider the circuit in *Fig. 1*. Derive the numerical expression of the transfer function

$$H(s) = \frac{V_o(s)}{V_g(s)}.$$

2. Using MATLAB, find the numerical value of each pole and zero of $H(s)$ and plot the pole-zero map.
3. Evaluate the inverse Laplace transform of $H(s)$ to find the impulse response $h(t)$. Verify your result using MATLAB. Create a suitable time vector and plot $h(t)$ against time. Obtain the impulse response using the `impulse` command as well and compare the plots.
4. Using the transfer function $H(s)$, find the output voltage $V_o(s)$ when $V_g(s)$ is a unit step function. Evaluate the inverse Laplace transform of $V_o(s)$ to find $v_o(t)$. Verify your result using MATLAB. Create a suitable time vector and plot $v_o(t)$ against time. Obtain the step response using the `step` command as well and compare the plots.

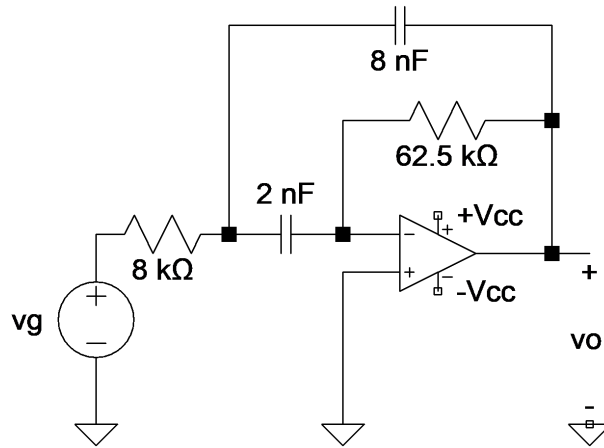


Figure 1

2 Answers

The s -domain equivalent circuit:

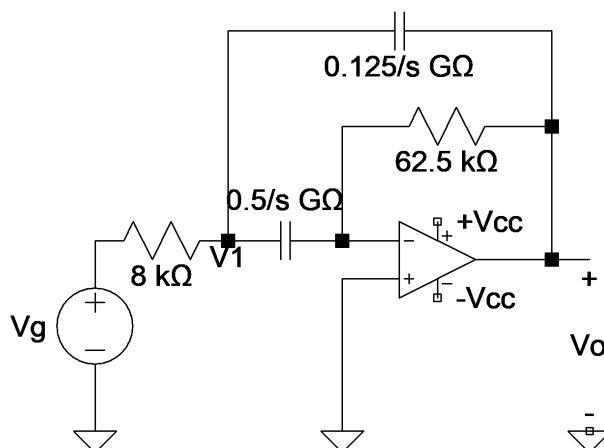


Figure 2

1. KCL at node V_1 :

$$\frac{V_1 - V_g}{8000} + \frac{sV_1}{5 \cdot 10^8} + \frac{(V_1 - V_0)s}{0.125 \cdot 10^9} = 0 \implies (625 \cdot 10^2 + 5s)V_1 - 4sV_0 = 625 \cdot 10^2 \cdot V_g \quad (1)$$

KCL at the negative terminal of the op-amp:

$$\frac{(0 - V_1)s}{5 \cdot 10^8} + \frac{0 - V_o}{62500} = 0 \implies V_1 = \frac{-8000}{s}V_o \quad (2)$$

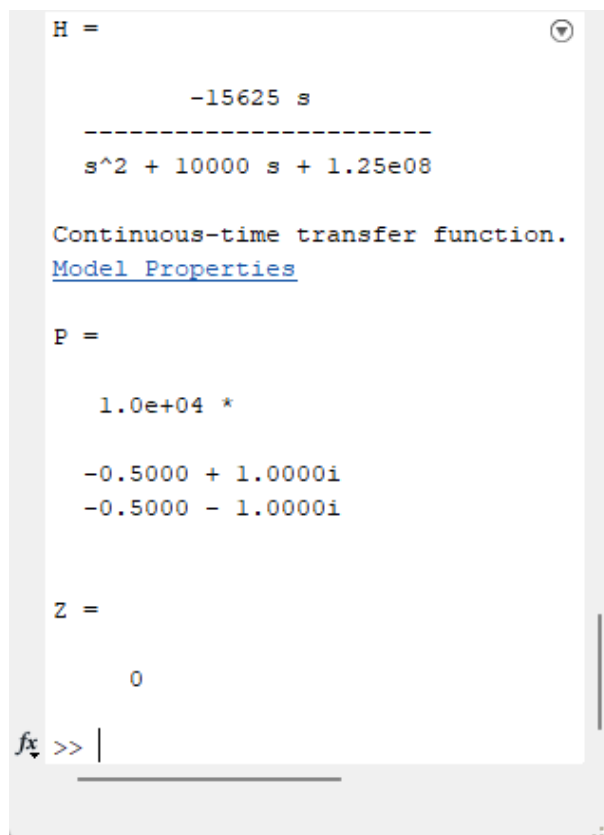
Substitute (2) into (1):

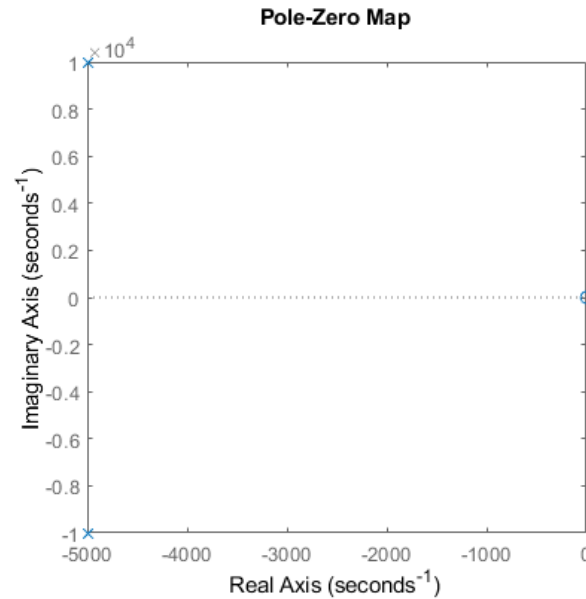
$$(62500 + 5s)\frac{-8000}{s}V_o - 4sV_o = 62500V_g \implies \frac{V_s}{V_g} = -\frac{15625s}{s^2 + 10000s + 125 \cdot 10^6}$$

2. Code:

```
num = [-15625 0];
denom = [1 10000 1.25e8];

H = tf(num, denom)
P = pole(H)
Z = zero(H)
pzmap(H)
```





3. $h(t)$ is the inverse Laplace transform of $H(s)$.

$$H = \frac{-15625s}{s^2 + 10000s + 1.25 \cdot 10^8} = \frac{-15625s}{(s + 5000 - j10000)(s + 5000 + j10000)}$$

$$H = \frac{K_1}{s + 5000 - j10000} + \frac{K_1^*}{s + 5000 + j10000}$$

$$K_1 = H \cdot (s + 5000 - j10000) \Big|_{s=-5000+j10000} = 8734.64 \angle -153.43^\circ$$

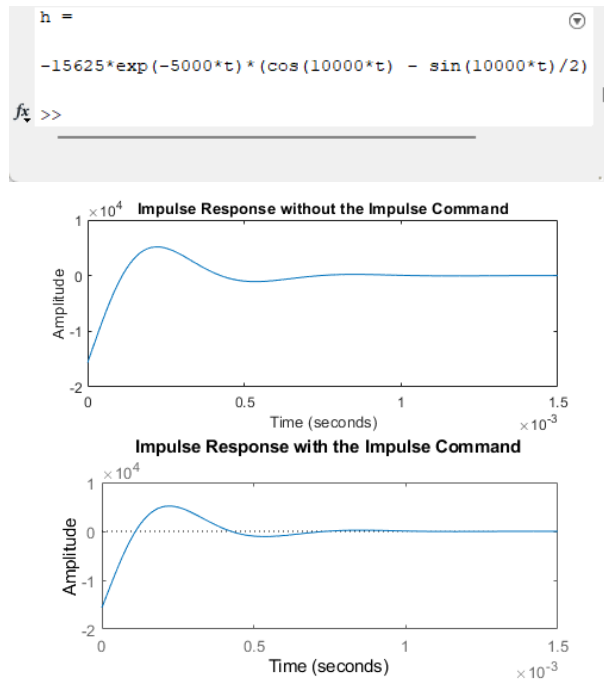
$$\Rightarrow \boxed{h(t) = [17469.28e^{-5000t} \cos(10000t - 153.43^\circ)]u(t)}$$

Code:

```
syms s t;
H_sym = poly2sym(num, s) / poly2sym(denom, s);
h = ilaplace(H_sym)
time = [0:10^(-6):1.5*10^(-3)];
figure

subplot(2, 1, 1);
h_vals = subs(h, t, time);
plot(time, h_vals)
xlabel('Time (seconds)')
ylabel('Amplitude')
title('Impulse Response without the Impulse Command')

subplot(2, 1, 2);
impulse(H, time)
title('Impulse Response with the Impulse Command')
```



4. Obtain the Laplace transform of the unit step function.

$$\mathcal{L}\{u(t)\} = \frac{1}{s} = V_g \implies V_o(s) = \frac{-15625s}{s^2 + 10000s + 0.125 \cdot 10^9}$$

$$V_o = \frac{K_1}{s + 5000 - j10000} + \frac{K_1^*}{s + 5000 + j10000}$$

$$K_1 = V_o \cdot (s + 5000 - j10000) \Big|_{s=-5000+j10000} = 0.78 \angle 90^\circ$$

$$\implies v_o(t) = [1.56e^{-5000t} \cos(10000t + 90^\circ)]u(t) \text{ V}$$

Code:

```
V_g = 1/s;
V_o = V_g * H_sym;
v_o = ilaplace(V_o)
figure

subplot(2, 1, 1);
v_o_vals = subs(v_o, t, time);
plot(time, v_o_vals)
xlabel('Time (seconds)')
ylabel('Amplitude')
title('Step Response without the Step Command')

subplot(2, 1, 2);
step(H, time)
title('Step Response with the Step Command')
```

```
v_o =  
-(25*sin(10000*t)*exp(-5000*t))/16  
fx >>
```

